Juliette Bruce's Scholarship, Teaching, Service

At the risk of being FINISH... A single conviction runs through much of my research, teaching, and service: *a desire to empower indiciduals to Balh balh by helping make hidden structers visibilte.*

Scholarship. My research is in algebraic geometry and commutative algebra FINISH. As described i my future reserch plan, my work follows three connected threads.

Regularity and Syzygies My doctorlal work studied the relations among systems of polynomial equations, often called syzyies, that describe higher dimensional shapes.

Tropical Geometry of Moduli Spaces

Since joining Dartmouth in Fall 2024 I have posted five new pre-prints, all of which are underreview at peer-reviewed juornals. My paper characterizing multgiraded regularity was accepted in *Advances in Mathematics*

Teaching and Mentoring. My first experiences teaching were as an undergraduate course assistant in inquiry-based courses at the University ofM Michigan. In these courses the students worked toegether to build and discover mathematics for themselves through crefully currated mix of group work, lecture, and carefully choosen problesm. That experience still shapes much of my teaching. I have attended six invitation research workshops since 2024 including two at Oberwolfach, the Banff International Reserarch Stateion, and the Americna Insitute of Mathematics. I was a longer term visintg research at the Fields Institue for a themeatic program in Spring 2025. COUNT TALKS

My advanced courses re where teaching and research meet most directly. Math 125 (Fal

Mentoring students is one of the parts of my job I find most rewarding. Currently I m advising two Ph.D. students: Will DeGroot (4th year) and Jacob Lehmann Duke (3rd year). ADD STUFF. I expect that within the year I will become Alex Moon's (2nd year) Ph.D. advisor as I hve been helping them preper for their advamcent exam. Beginning in Summer 2026 I am mentoing a post-doc Amy Li who finished her Ph.D at the UT Ausint. TOGETHER JACOB¡ AELX AND AMY AND I ARE WORKING ON BT STUFF

Beyond Dartmouth, in summer 2026 I led four undergraduates on a summer search program through *Count Me In* held at the University of Edinburgh.

Service and Community. Mathematics is a deeply collborative feild, and much of its culture is passed along informally.

At Dartmouth. In fall 2025 I founded *Unwritten Rules*, a graduate seminar that seeks to make the hidden curriculum and expectation of an academic career explicitly. Since its founding ADD THINGS WE MEET 1-2 TIMES A MONTH (CONFERENES, FUNDING. JOURNALS). Since 2025 I serve as the faculty co-advisor for Dartmouth's student chapter of the Association for Women in Mathematics. DINNER ON GOING TO GRAD SCHOOL. In November 2024 I organized the Algebraic Geometry Northeastern Series (AGNES) and BATMOBILE conferences at Dartmouth, which brought over 150 mathamticians to Dartmouth to discuss recent breakthrohgs in algebraci geometry. Currently, I am in the early planning stages of organaing *GEMS of Commutative Algebra*, a conference building research networks for early-career researchers in commuttive algebra, which I hope to held at Dartmouth in 2028.

In the profession. I was the inaugural president for *Spectra, the Association for LGBTQ+ Mathematicians*, and led the adoption of bylaws,the creation of a invited lecture series at the Joint Mathematics Meetings, and fundraised over $20,000 in seed funding. I am currently serving on the board of trustees for *Spectra* working to guide the organiziation to longterm ADD PUFF. From 2023 to 2025 I served on SLMath's Committee on Women in Mathematics. Currently I serve on the American Mathematical Sociey's Committee on Engagement, Participation, and Advacnement and the Committee on AMS-Simons Travel grants. I have reviewed grants for the National Science Foundation, refereeed for more than a dozen different journals, and ADD NUMBER OF CONFERECNES WITH TOTAL PEOPLE.